

Module 2: Management of Environmental Problems

EE 8308 Environmental Engineering Management

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Learning Outcomes

- Explain the purpose and elements of an Environmental Management System (EMS) based on ISO 14001.
- Develop and use Key Performance Indicators (KPIs) for energy, water, waste, and materials.
- Calculate improvements in energy and water intensity.
- Conduct environmental audits and compute compliance scores.
- Understand management review inputs and outputs.

What is an Environmental Management System?

Definition

An EMS is part of an organization's management system used to manage environmental aspects, fulfill compliance obligations, and address risks and opportunities.

Why EMS for Electrical and Electronic Engineering (EEE)?

- **Legal compliance:** WEEE, RoHS, local environmental laws.
- **Cost reduction:** Energy, water, waste, materials.
- **Market access:** ISO 14001 certification often required by large buyers.
- **Risk management:** Prevent fines, spills, reputational damage.

Required Elements of ISO 14001

Key requirements

- Appropriate to purpose, size, nature of impacts.
- Commitment to comply with legal requirements.
- Commitment to pollution prevention and continual improvement.
- Framework for setting environmental objectives.
- Documented, communicated, available to public.

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Example policy excerpt (EEE manufacturer):

"We are committed to reducing our product carbon footprint by 30% by 2030, achieving zero waste to landfill by 2025, and ensuring all suppliers comply with RoHS and REACH. We will regularly report progress to stakeholders."

Key Performance Indicators (KPIs) for EEE

Category	KPI	Unit
Energy	Energy intensity	kWh per unit product
Carbon	CO ₂ -eq intensity	kg CO ₂ -eq per unit product
Water	Water intensity	m ³ per unit product
Waste	Waste diversion rate	% (recycled + reused)
Materials	Recycled content	% of total material mass
Compliance	Non-conformities	number per audit
Supplier	Green supplier score	% spend with certified suppliers

Formula for Energy Intensity

Energy Intensity (EI)

$$EI = \frac{E_{\text{total}}}{\text{Production volume}} \quad [\text{kWh/unit}]$$

Used to track improvement over time and benchmark against competitors. Similarly for water intensity:

$$WI = \frac{\text{Total water use}}{\text{Production volume}} \quad [\text{m}^3/\text{unit}]$$

Worked Example 3 – Energy Intensity Improvement

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Step 4: Annual energy cost saving (electricity USD 0.12/kWh)

$$\text{Saving} = (500,000 - 450,000) \times 0.12 = 50,000 \times 0.12 = 6,000 \text{ USD/year}$$

Practice Problem 3 – Water Intensity

Problem

An electrical cable manufacturer produces 2000 km of cable per year.

- 1 Current water use: $8000 \text{ m}^3/\text{year}$. Calculate water intensity (m^3/km).
- 2 After recycling cooling water, water use drops to $6000 \text{ m}^3/\text{year}$. What is the reduction in water intensity?
- 3 If water cost is TZS $2000/\text{m}^3$, what is the annual cost saving?
- 4 Express the saving in USD ($1 \text{ USD} = 2500 \text{ TZS}$).

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- ② After recycling cooling water, water use drops to 6000 m³/year. What is the reduction in water intensity?
- ③ If water cost is TZS 2000/m³, what is the annual cost saving?
- ④ Express the saving in USD (1 USD = 2500 TZS).

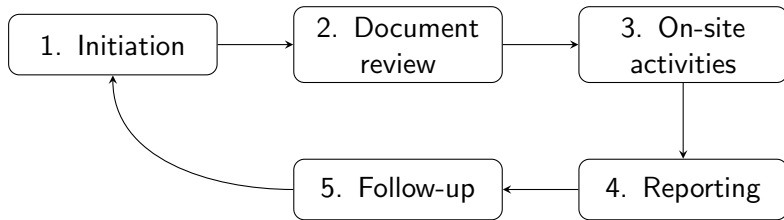
Solution:

- ① $WI_{old} = 8000/2000 = 4 \text{ m}^3/\text{km}$.
- ② $WI_{new} = 6000/2000 = 3 \text{ m}^3/\text{km}$. Reduction = 1 m³/km.
- ③ Saving = $(8000 - 6000) \times 2000 = 2000 \times 2000 = 4,000,000 \text{ TZS/year}$.
- ④ In USD: $4,000,000/2500 = 1600 \text{ USD/year}$.

Types of Environmental Audits

- **Compliance audit:** Verify adherence to legal and regulatory requirements.
- **Performance audit:** Assess effectiveness of EMS and progress toward objectives.
- **Supplier audit:** Evaluate environmental performance of supply chain.
- **Life cycle audit:** Examine all stages from raw material to end-of-life.
- **Merger/acquisition audit:** Identify environmental liabilities.

Audit Process (ISO 19011)



- **Initiation:** scope, objectives, team.
- **Document review:** policies, records, permits.
- **On-site:** interviews, observations, sampling.
- **Reporting:** findings, non-conformities, opportunities.
- **Follow-up:** corrective actions, verification.

Worked Example 4 – Audit Non-Conformity Calculation

Problem: During an EMS audit, the following non-conformities were found. Calculate the compliance score.

Type	Count	Weight (points each)
Major non-conformities (must fix immediately)	2	5
Minor non-conformities (fix within 30 days)	5	2
Opportunities for improvement (OFIs)	8	0

Maximum possible score (zero non-conformities) = 100.

Worked Example 4 – Audit Non-Conformity Calculation

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Step 1: Deduction points

$$\text{Deductions} = 2 \times 5 + 5 \times 2 = 10 + 10 = 20$$

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Step 2: Compliance score

$$\text{Score} = 100 - 20 = 80\%$$

Practice Problem 4 – Transformer Plant Audit

Problem

An audit of a transformer manufacturing plant reveals:

- **Major:** missing emergency response plan, no monitoring of oil spills.
 - **Minor:** two incomplete training records, missing calibration of pH meter.
 - **OFls:** improve lighting efficiency, install leak detection.
- ① Count majors, minors, OFIs.
 - ② Calculate compliance score using weighting: major = 10 pts, minor = 3 pts, OFI = 0.
 - ③ If the plant needs a score of 85% to keep ISO 14001, does it pass?
 - ④ Suggest corrective actions for each major non-conformity.

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Solution:

- ① Majors = 2, Minors = 2, OFIs = 2.
- ② Deductions = $2 \times 10 + 2 \times 3 = 20 + 6 = 26$. Score = $100 - 26 = 74\%$.

Management Review – Inputs and Outputs (ISO 14001:2015)

Inputs:

- Status of previous actions
- Changes in external/internal issues
- Environmental performance (KPIs, compliance)
- Adequacy of resources
- Effectiveness of controls
- Opportunities for improvement

Outputs:

- Decisions for continual improvement
- Changes to EMS (policy, objectives)
- Resource needs
- Actions if objectives not met

Frequency: Typically annually. For high-risk EEE operations (e.g., battery recycling), semi-annual recommended.

Summary – Module 2

- **EMS (ISO 14001):** Plan-Do-Check-Act cycle for environmental management.
- **KPIs:** Energy intensity, water intensity, waste diversion, etc.
- **Improvement calculation:** $\Delta\% = \frac{\text{old} - \text{new}}{\text{old}} \times 100\%$.
- **Audits:** Types, process, scoring (deductions from 100).
- **Management review:** Regular check of system effectiveness.

Good EMS reduces costs, risks, and environmental impact.

Thank You!

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